

Notes: Walker (QJE, 2013)
The Transitional Costs of Sectoral Reallocation: Evidence from the Clean Air Act and the Workforce

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1 Big picture

- **Research question.** What are the transitional labor-market costs when environmental regulation reallocates production away from newly regulated plants? In particular, how much do affected workers lose in earnings, employment continuity, and job matches after the 1990 Clean Air Act Amendments (CAAA)?
- **Main idea.** Counting “jobs lost” is not enough. A worker leaving a regulated plant may quickly find a similar job, may suffer a long nonemployment spell, or may move to a lower-paying sector. The policy-relevant cost is the worker’s earnings path, not just the sector’s employment level.
- **Empirical setting.** The 1990 CAAA newly classified many counties as nonattainment for particular pollutants. Only polluting plants in counties violating the relevant pollutant standard face the new regulatory burden.
- **Empirical object.** Walker follows workers who were employed in polluting plants before the regulation and compares their later earnings to comparable workers in nonpolluting sectors and/or unaffected counties.
- **Main finding.** Workers in newly regulated plants lose roughly 20% of one preregulatory year’s earnings in present discounted value over the following nine years. In aggregate, the forgone wage bill is about \$5.4 billion in 1990 dollars.
- **Mechanism.** The losses are not mainly wage cuts for workers who stay. They are driven by workers who separate from their original plants, spend time without recorded earnings, and often move to different industries.
- **Policy takeaway.** The estimated worker costs are meaningful for affected workers but small relative to the estimated health benefits of the 1990 CAAA. The paper therefore highlights distributional costs rather than overturning the case for environmental regulation.

2 Incremental contribution of the paper

- **From sector jobs to worker incidence.** Earlier work usually studies employment in regulated industries. Walker studies the same workers over time, which makes nonemployment and post-transition earnings losses visible.
- **Linked worker-firm evidence.** The paper uses confidential Longitudinal Employer Household Dynamics (LEHD) data to follow workers across firms, industries, locations, and years.
- **Plant-level regulation.** Instead of proxying regulation with broad industry pollution intensity, the paper links Census plant data to EPA Air Facility Subsystem permits to identify which plants are actually regulated.
- **Clean triple-difference design.** The regulatory shock varies by county, pollutant, time, and plant polluter status. This supports a difference-in-difference-in-differences design.
- **Cohort approach.** By fixing cohorts based on the 1990 job, the paper avoids the composition problem that arises when industry averages change because lower-wage or higher-wage workers leave.

- **Transition mechanisms.** The paper separates stayers from leavers, identifies industry and county destinations, and estimates the contribution of nonemployment.
- **General lesson.** The framework is useful beyond environmental policy: trade shocks, local labor-demand shocks, energy shocks, and other sectoral reallocations can have similar transitional incidence.

3 Data and measurement

3.1 Clean Air Act regulatory variation

The Clean Air Act requires the EPA to set National Ambient Air Quality Standards (NAAQS) for major air pollutants. Counties that violate a standard are designated as nonattainment counties. In those counties, polluting sources face tighter requirements, including more stringent abatement technology and offset requirements.

- **County variation.** Some counties are newly designated as nonattainment after the 1990 CAAA.
- **Pollutant variation.** A county can be nonattainment for one pollutant but not another.
- **Plant variation.** Only plants emitting the pollutant for which the county is in violation are directly regulated.
- **Timing.** The key policy shock occurs around the 1991 implementation of the 1990 CAAA.

Interpretation. The design compares workers in polluting plants hit by a new regulation to workers in similar places or sectors not hit by that specific regulation.

3.2 LEHD worker data

The primary worker-level data source is the U.S. Census Bureau’s LEHD file, which contains administrative quarterly earnings records for a large share of the U.S. workforce.

- The baseline sample begins with workers in manufacturing and electric/gas industries in 1990.
- The available pre-1991 LEHD states are Illinois, Maryland, Washington, and Wisconsin.
- The paper follows roughly 3 million workers over the next 10 years.
- Earnings are converted to annual values and deflated to 1990 dollars.
- Missing earnings are interpreted carefully because they can reflect nonemployment, nonparticipation, or movement outside covered states.

Interpretation. LEHD makes it possible to include workers who leave the regulated sector, rather than measuring only wages among workers still observed in the sector.

3.3 LBD plant data and EPA AFS regulation data

Walker uses the Longitudinal Business Database (LBD) to measure plant-level employment, payroll, age, industry, location, and entry/exit. The LBD is also useful for checking pre-trends and for measuring sectoral employment responses.

The EPA Air Facility Subsystem (AFS) identifies plants with air permits and their regulated pollutants. Walker matches AFS plants to Census establishments using names and addresses.

- A plant is treated as regulated if it has an EPA air permit and is located in a county that is nonattainment for the pollutant it emits.
- The paper focuses mainly on PM₁₀ and ozone-related designations because these generate the main regulatory variation.
- Ozone-emitting plants are identified by emissions of ozone precursors such as VOCs and NO_x.

3.4 Cohort construction and outcomes

For the worker analysis, cohorts are defined by the worker’s 1990 county, industry, and polluting sector. A sector is one of four categories: PM₁₀ polluter, ozone precursor polluter, both, or neither.

Main outcomes include:

- mean log earnings of the 1990 cohort in each later year;
- separation from the 1990 plant;
- destination of job transition by industry and county;
- time with missing earnings, interpreted as nonemployment or nonparticipation;
- distributional position in the preregulatory earnings distribution.

Interpretation. Cohorts hold fixed the original set of workers. This is crucial because the composition of workers still employed in a sector can change after regulation.

4 Methodology and empirical specifications

4.1 Baseline triple-difference specification

The treatment indicator combines county nonattainment status, plant or sector polluter status, and post-regulation timing:

$$Y_{jcst} = \beta_1 [N_c \times P_s \times \mathbf{1}(t > 0)] + \alpha_{jcs} + \nu_{ct} + \pi_{st} + \lambda_{jt} + \varepsilon_{jcst}.$$

Here Y_{jcst} is an outcome for industry j , county c , sector s , and year t ; N_c indicates a newly nonattainment county for a specific pollutant; P_s indicates a sector that emits that pollutant; and $\mathbf{1}(t > 0)$ marks post-regulation years.

The fixed effects absorb:

- time-invariant county-industry-sector differences;
- shocks common to newly nonattainment counties in a year;
- shocks common to polluting sectors in a year;
- industry-year shocks.

4.2 Event-study version

The main estimates use a dynamic event-study specification:

$$Y_{jct} = \sum_{k=-m}^M \beta_k [N_c \times P_s \times \mathbf{1}(t = k)] + \alpha_{jcs} + \nu_{ct} + \pi_{st} + \lambda_{jt} + \varepsilon_{jct}.$$

The coefficients β_k show the time path of outcomes before and after regulation. The year before treatment is the omitted benchmark.

4.3 Sectoral employment analysis

The paper first estimates whether regulation causes sectoral downsizing. The dependent variable is log employment in a county-sector-year. This establishes that the policy shock actually reallocates employment away from treated polluting sectors.

4.4 Cohort earnings analysis

The central outcome is mean log earnings for the 1990 worker cohort. Because the cohort is fixed, the estimate captures both:

- earnings changes for workers who stay at the original regulated plant; and
- earnings losses for workers who leave, including nonemployment and lower pay in future jobs.

4.5 Stayers, leavers, and job destinations

Walker decomposes earnings effects by whether workers stay at the original plant or separate within the first four years. For separators, the paper further separates destinations:

- same industry, same county;
- different industry, same county;
- same industry, different county;
- different industry, different county.

Interpretation. This decomposition reveals whether losses are mostly due to job loss itself, loss of industry-specific human capital, geographic relocation, or local labor-market conditions.

5 Identification and interpretation

5.1 What the paper can identify

The paper estimates the reduced-form effect of new plant-level environmental regulation on the earnings paths of workers employed in treated sectors just before the regulation. It is best interpreted as the transitional cost borne by directly exposed workers, not as the full general-equilibrium effect of environmental policy.

5.2 Why the 1990 CAAA helps identification

The 1990 CAAA created a discrete change in regulatory status for many counties. This is useful because ordinary nonattainment status can be correlated with local economic activity and pollution levels. The discrete policy shock creates more credible before-after variation.

5.3 Why plant-level polluter status matters

Within a newly nonattainment county, only plants emitting the relevant pollutant are regulated. This allows comparisons between polluting and nonpolluting sectors within similar local labor markets.

5.4 Composition-bias concern

Industry wage averages can be misleading after a shock: if low-wage workers leave, the average wage of remaining workers may not fall even if displaced workers suffer large losses. Walker addresses this by following fixed worker cohorts.

5.5 Spillover concern

Workers leaving regulated plants may enter nearby sectors, potentially affecting the control group. Walker discusses local labor-market spillovers and shows that the estimates are somewhat attenuated under stricter within-county comparisons, but the main pattern remains.

5.6 External-validity concern

The LEHD sample starts with four states that have pre-1991 data. These states account for a meaningful but incomplete share of newly regulated polluting-sector employment. The paper extrapolates carefully to national aggregate earnings losses.

6 Tables and figures

6.1 Table I and Figure I: sample characteristics and the source of regulatory variation

Read Table I:

- Polluting plants are larger and older than nonpolluting plants.
- Workers in polluting plants earn substantially more than workers in nonpolluting plants.
- Separation rates are lower in polluting plants before regulation, consistent with more stable jobs.
- Newly nonattainment counties have sizable polluting-sector employment and wage bills.

Read Figure I:

- Panel (a) maps all counties in nonattainment for any pollutant in 1991.
- Panel (b) isolates counties newly entering nonattainment in 1991.
- New nonattainment counties are not limited to one region, although regulation is concentrated in industrial and metropolitan areas.

Economic message: The paper’s identification is not simply “dirty counties versus clean counties.” It combines new county-level regulation with plant-level polluter status.

TABLE I
SAMPLE STATISTICS: PREREGULATION (1990)

	(1) Attainment	(2)	(3) Nonattainment	(4)	(5) Switch into nonattainment	(6)
	Nonpolluting	Polluting	Nonpolluting	Polluting	Nonpolluting	Polluting
Panel A: Plant characteristics (LRD)						
No. of plants	6,896	1,797	5,910	1,549	7,622	1,452
Plant employment	146.82	244.6	133.15	266.42	144.24	234.58
Plant age	8.64	10.07	8.4	10.89	8.84	10.71
Panel B: Workforce characteristics (LEHD)						
Age	36.88	38.61	36.67	38.3	37.04	38.1
Education	13.34	13.56	13.72	14.08	13.67	13.89
Earnings	26,996.56	36,950.09	31,159.89	38,030.07	31,692.56	39,307.85
Separation rate	0.22	0.08	0.24	0.11	0.23	0.08
Same ind. same county	0.04	0.01	0.05	0.02	0.04	0.01
Same ind. diff county	0.02	0.01	0.05	0.02	0.01	0.00
Diff ind. same county	0.06	0.02	0.05	0.02	0.1	0.03
Diff ind. diff county	0.09	0.03	0.08	0.03	0.06	0.02
Leave sample	0.01	0.01	0.02	0.02	0.02	0.02
Seg. job to job	0.39	0.06	0.2	0.08	0.18	0.05
Panel C: Sectoral characteristics (LRD and LEHD)						
Total employment	233,994	216,386	179,590	146,123	379,553	299,364
Total earnings (in \$millions)	6,317.03	7,773.78	5,596	5,557.07	12,029.01	8,229.67

Note: Sample statistics for the "base" states from which the LEHD workforce sample is drawn (IL, MD, WA, WI). Since these are statistics for the "base" year, they only include the manufacturing industry. All dollar amounts are reported in 1990 dollars. Separation is defined as working in 1990 and working either at a different plant in 1991 or having missing earnings in 1991. The separation rate is further decomposed into four different categories listed in the table. LEHD denotes characteristics derived from the LEHD sample. LEHD denotes characteristics from the LEHD worker cohort sample.

Table I: prereregulation sample statistics.

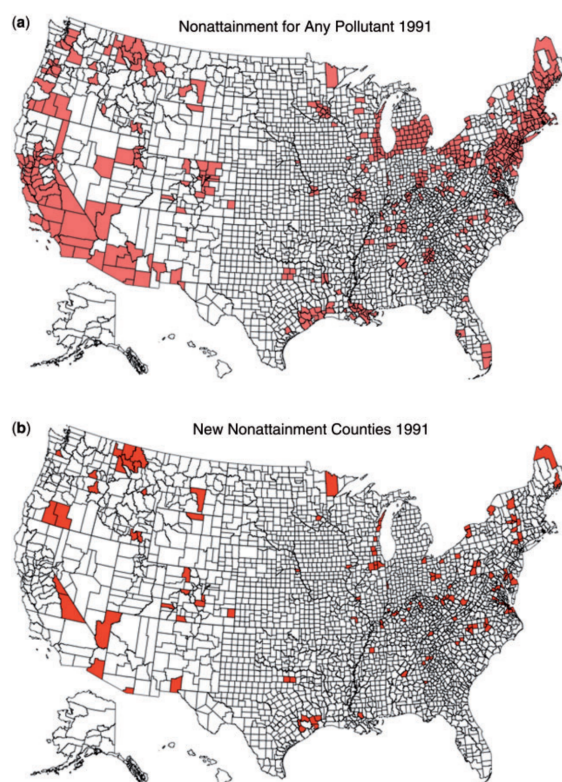


Figure I: nonattainment counties and newly regulated counties in 1991.

6.2 Figure II and Table II: sectoral employment and main earnings effects

Read Figure II:

- Employment in newly regulated polluting sectors falls after the CAAA shock.
- Pre-trends are relatively flat, supporting the event-study design.
- The employment decline reaches roughly 10% below 1990 levels and about 15% below the counterfactual path after several years.

Read Table II:

- Cohort earnings fall immediately after regulation.
- The loss is largest in the first few years and gradually recovers after about five years.
- The preferred estimates imply a nine-year present discounted earnings loss of roughly 16% to 24% of prereregulatory annual earnings, depending on controls.
- The baseline interpretation is about 20% of one year's prereregulatory earnings.

Economic message: Regulation causes real sectoral downsizing, and worker earnings losses persist even after workers have time to adjust.

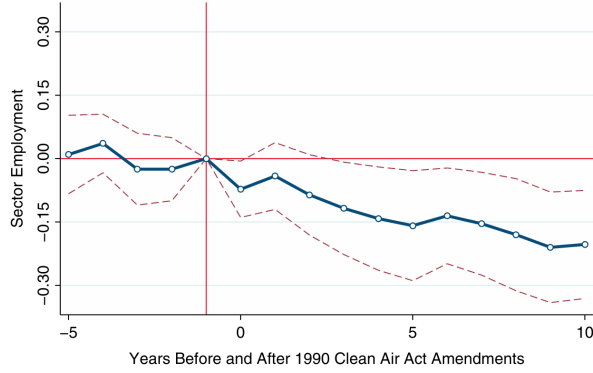


Figure II: sector employment falls after regulation.

TABLE II
EFFECT OF SECTOR-LEVEL REGULATION ON EARNINGS

	(1)	(2)	(3)	(4)	(5)	(6)
Regulation ($t+0$)	-0.033** (0.014)	-0.031** (0.012)	-0.034** (0.017)	-0.036** (0.015)	-0.036** (0.017)	-0.033*** (0.010)
Regulation ($t+1$)	-0.058*** (0.012)	-0.056*** (0.014)	-0.057*** (0.019)	-0.059*** (0.011)	-0.056*** (0.014)	-0.051*** (0.012)
Regulation ($t+2$)	-0.046*** (0.012)	-0.045*** (0.011)	-0.062*** (0.009)	-0.040*** (0.009)	-0.051*** (0.010)	-0.030** (0.012)
Regulation ($t+3$)	-0.036** (0.017)	-0.034** (0.016)	-0.048* (0.026)	-0.028** (0.012)	-0.035** (0.016)	-0.019** (0.009)
Regulation ($t+4$)	-0.041 (0.026)	-0.040 (0.025)	-0.054 (0.033)	-0.034** (0.016)	-0.040** (0.019)	-0.019** (0.008)
Regulation ($t+5$)	-0.011 (0.014)	-0.010 (0.015)	-0.020** (0.009)	-0.013 (0.014)	-0.015 (0.009)	-0.011 (0.014)
Regulation ($t+6$)	0.000 (0.016)	0.001 (0.017)	-0.002 (0.012)	-0.003 (0.012)	0.001 (0.009)	-0.011* (0.006)
Regulation ($t+7$)	0.003 (0.012)	0.004 (0.012)	0.008 (0.013)	-0.004 (0.011)	0.007 (0.012)	-0.010 (0.009)
Regulation ($t+8$)	0.005 (0.010)	0.006 (0.010)	0.009 (0.008)	0.001 (0.011)	0.004 (0.009)	0.008 (0.008)
9-year PDV	-0.202*** (0.047)	-0.191*** (0.046)	-0.241*** (0.050)	-0.199*** (0.044)	-0.204*** (0.044)	-0.162*** (0.054)
N	153,249	153,249	153,249	153,249	153,249	153,249
2-digit SIC $\times$ year FE				X	X	
County trends		X		X		
County $\times$ year FE			X		X	
County $\times$ SIC $\times$ year FE						X

Note. This table reports regression coefficients from six separate regressions based on equation (2) where the dependent variable consists of mean log earnings in a cohort $\times$ year. Cohorts are defined by the county $\times$ industry $\times$ sector of work in 1990, and a sector is defined by one of four polluting categories. See text for details. All of the regressions implicitly or explicitly control for cohort specific FE, nonattainment $\times$ year FE, polluting sector $\times$ year FE, the fraction of workers in various potential experience bins (<5, 5–8, 8–13, 13–22, 22–32, 32–42, 42–47, and 47–55), quadratic trends in average cohort education, quadratic trends in county-level demographics, and year fixed effects. Standard errors are in parentheses and are clustered by commuting zone. Exponentiated coefficients are reported using the translation $(\exp(\beta_i) - 1)$. The regressions are weighted by the 1990 cohort size. The final row of the table reports the discounted sum of the coefficients using a 4% annual discount rate (with discount factor β) and the translation $\sum_{t=0}^8 \beta^t (\exp(\beta_i^t) - 1)$. Standard errors for the final row are calculated using the delta method. ***, **, * denotes statistical significance at the 1%, 5%, and 10% levels respectively.

Source: LEHD.

Table II: main cohort earnings effects.

6.3 Figure III and Table III: earnings dynamics, stayers, and leavers

Read Figure III:

- Cohort earnings fall sharply after nonattainment designation.
- The shaded area represents the forgone wage bill over the post-regulation years.
- Earnings recover gradually but do not immediately return to the pre-regulation path.

Read Table III:

- Stayers experience little or no cumulative earnings loss.
- Separators experience very large losses: the nine-year PDV is more than one full preregulatory year's earnings.
- Workers who switch industries lose more than workers who remain in the same industry.
- The evidence points to costly job transitions and loss of industry-specific human capital.

Economic message: The transitional cost is not primarily a wage cut inside surviving regulated plants. It is the cost of being reallocated out of those plants.

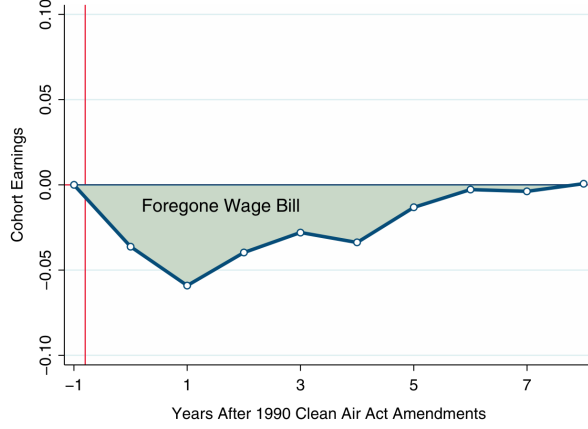


Figure III: cohort wage trends and the foregone wage bill.

TABLE III
EFFECT OF SECTOR-LEVEL REGULATION ON EARNINGS DECOMPOSITION BASED ON SECTORAL TRANSITIONS

	(1)	(2)	(3)	(4)	(5)	(6)
	Stayer	Separator	Separator: same industry same county	Separator: diff. industry same county	Separator: same industry diff. county	Separator: diff. industry diff. county
Regulation (t + 0)	-0.011 (0.019)	-0.087*** (0.007)	-0.023 (0.021)	-0.084*** (0.011)	-0.125*** (0.015)	-0.084*** (0.008)
Regulation (t + 1)	-0.027** (0.012)	-0.184*** (0.011)	-0.123*** (0.012)	-0.171*** (0.011)	-0.124*** (0.022)	-0.178*** (0.012)
Regulation (t + 2)	0.004 (0.009)	-0.265*** (0.026)	-0.156*** (0.026)	-0.235*** (0.022)	-0.174*** (0.012)	-0.258*** (0.022)
Regulation (t + 3)	0.004 (0.012)	-0.267*** (0.030)	-0.220*** (0.030)	-0.257*** (0.040)	-0.179*** (0.029)	-0.272*** (0.029)
Regulation (t + 4)	-0.008 (0.018)	-0.208*** (0.036)	-0.153*** (0.036)	-0.186*** (0.045)	-0.109*** (0.020)	-0.225*** (0.022)
Regulation (t + 5)	0.014 (0.015)	-0.168*** (0.021)	-0.136*** (0.021)	-0.169*** (0.029)	-0.088*** (0.016)	-0.174*** (0.013)
Regulation (t + 6)	0.011 (0.011)	-0.113*** (0.011)	-0.082 (0.016)	-0.107*** (0.012)	-0.032 (0.021)	-0.138*** (0.013)
Regulation (t + 7)	0.006 (0.012)	-0.063*** (0.010)	-0.026 (0.017)	-0.060*** (0.011)	0.004 (0.014)	-0.056*** (0.010)
Regulation (t + 8)	0.007 (0.016)	-0.034** (0.014)	-0.002 (0.010)	-0.030 (0.020)	-0.014* (0.008)	-0.040*** (0.007)
9-year PDV	-0.000 (0.053)	-1.225*** (0.098)	-0.810*** (0.141)	-1.105*** (0.120)	-0.770*** (0.067)	-1.244*** (0.082)
N	152,888	153,160	151,523	152,715	151,929	153,025

Note: This table reports regression coefficients from equation (2). Each column corresponds to a different regression, where the results are presented for six different samples as indicated in the table headings. An observation is a cohort $\times$ year where cohorts are defined by the county $\times$ industry $\times$ sector of work in 1990, and a sector is defined by one of four polluting categories. See text for details. All of the regressions control for cohort specific FE, nonattainment $\times$ cohort FE, polluting sector $\times$ year FE, county-specific time trends, the fraction of workers in various potential exposure bins (3, 5, 8, 9, 11, 13–22, 23–32, 33–42, 43–47, and 47–55), quadratic trends in average cohort education, quadratic trends in county-level demographics, and industry-year fixed effects. Standard errors are in parentheses and are clustered by commuting zone. Exponentiated coefficients are reported using the translation $\exp(\cdot) - 1$. The regressions are weighted by the 1990 cohort size. The final row of the table reports the discounted sum of the coefficients using a 4% annual discount rate (with discount factor β and the translation $1 - \beta / \exp(r) = 1$). Standard errors for the final row are calculated using the delta method. ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Source: LEED.

Table III: decomposition for stayers, separators, and destination sectors.

6.4 Figure IV and Table IV: destination sectors and heterogeneous incidence

Read Figure IV:

- Earnings losses are largest for workers who move to different industries.
- Same-industry transitions are less costly than different-industry transitions.
- Location also matters, but industry switching is the more important margin in the plotted estimates.

Read Table IV:

- Older workers suffer larger percentage losses than younger workers.
- Women and high-initial-earnings workers experience larger losses as a share of preregulatory income.
- Losses are much larger in high-unemployment counties, showing that macro and local labor-market tightness matter.
- PM₁₀ regulation appears to have a larger effect than ozone regulation, though the difference is not always statistically decisive.

Economic message: Reallocation costs are not uniform. They are concentrated among workers with more job-specific rents and in weaker local labor markets.

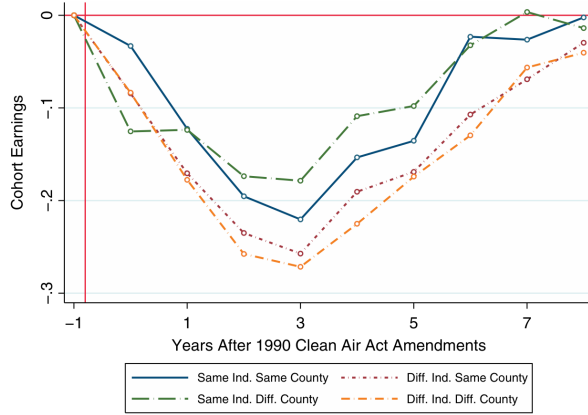


FIGURE IV

Figure IV: earnings losses by transition destination.

TABLE IV
HETEROGENEOUS EFFECTS OF SECTOR-LEVEL REGULATION ON COHORT EARNINGS

Panel A: Age				
	(1) Prime-age male workers	(2) 25 ≤ age < 35	(3) 35 ≤ age < 45	(4) 45 ≤ age < 55
9-year PDV	-0.212*** (0.060)	-0.113** (0.051)	-0.216*** (0.055)	-0.430*** (0.108)
N	124,671	125,959	112,792	90,280
Panel B: Worker sex		Panel C: Initial worker earnings		
	(1) Male	(2) Female	(1) Initial earnings < firm average	(2) Initial earnings ≥ firm average
9-year PDV	-0.154*** (0.047)	-0.271*** (0.048)	-0.080* (0.046)	-0.270*** (0.045)
N	138,829	124,266	122,573	150,211
Panel D: Initial plant earnings		Panel E: Initial plant size		
	(1) Avg. plant earnings > avg. sector earnings	(2) Avg. plant earnings ≤ avg. sector earnings	(1) Plant emp ≤ 150	(2) Plant emp > 150
9-year PDV	-0.185 (0.255)	-0.188** (0.081)	-0.039 (0.184)	-0.321*** (0.108)
N	118,889	119,681	125,991	88,083
Panel F: Unemployment rate		Panel G: Pollutant-specific		
	(1) Below median	(2) Above median	(1) PM10	(2) Ozone
9-year PDV	-0.062 (0.119)	-0.258*** (0.086)	-0.198*** (0.061)	-0.138 (0.107)
N	87,636	65,613	153,249	

Note. This table reports regression coefficients from 15 separate regressions based on equation (2) where the dependent variable consists of mean log earnings in a cohort $\times$ year. Each panel presents two separate regressions, with the exception of Panel G, which reports the results from a single regression. Cohorts are defined by the county $\times$ industry $\times$ sector of work in 1990, and a sector is defined by one of four polluting categories. See text for details. All of the regressions control for cohort specific FE, non-attainment $\times$ year FE, polluting sector $\times$ year FE, county-specific time trends, the fraction of workers in various potential experience bins (<5, 5-8, 8-13, 13-22, 22-32, 32-42, 42-47, and 47-55), quadratic trends in average cohort education, quadratic trends in county-level demographics, and industry-year fixed effects. The table reports the discounted sum of event-time coefficients using a 4% annual discount rate (with discount factor β) and the translation $\sum_{j=0}^{\infty} \beta^j (\exp(\eta_j^2) - 1)$. The regressions are weighted by the 1990 cohort size. Standard errors are calculated using the delta method and are clustered by commuting zone. ***, **, * denotes statistical significance at the 1%, 5%, and 10% levels, respectively.
Source. LEHD.

Table IV: heterogeneity by worker, plant, labor-market, and pollutant characteristics.

6.5 Table V and Figure V: distributional effects and separations

Read Table V:

- Regulation increases the fraction of workers at the very bottom of the preregulatory earnings distribution.
- It reduces the fraction of workers at the top of the earnings distribution.
- The middle of the earnings distribution is less affected.

Read Figure V:

- Separation rates from the 1990 plant rise after regulation.
- The separation effect peaks several years after the policy shock.
- The bars show that the share of workers still at the original plant steadily declines over time.

Economic message: The earnings losses are generated by a shift from high or stable preregulatory jobs toward nonemployment or lower-ranked earnings outcomes.

TABLE V
EFFECT OF SECTOR LEVEL REGULATION ON THE EARNINGS DISTRIBUTION

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	$0 \leq p \leq 1$	$1 < p \leq 5$	$5 < p \leq 10$	$10 < p \leq 25$	$25 < p \leq 50$	$50 \leq p \leq 75$	$75 < p \leq 90$	$90 < p \leq 95$	$95 < p \leq 99$	$99 < p \leq 100$
9-year total	0.045*	-0.005	-0.009	0.013	0.061	0.016	-0.005	-0.024	-0.064***	-0.028***
	(0.025)	(0.017)	(0.006)	(0.030)	(0.063)	(0.033)	(0.051)	(0.028)	(0.026)	(0.009)
N	156,324	156,324	156,324	156,324	156,324	156,324	156,324	156,324	156,324	156,324

Note: This table reports regression coefficients from nine separate regressions based on equation (2) where the dependent variable consists of the fraction of workers in the pretreatment, cohort-specific earnings percentile in the years after the regulations. The earnings percentiles are indicated in the column headings. Cohorts are defined by the county $\times$ industry $\times$ sector of work in 1990, and a sector is defined by one of four polluting categories. See text for details. All of the regressions control for cohort specific FE, treatment $\times$ year FE, polluting sector $\times$ year FE, county-specific time trends, the fraction of workers in various potential experience bins (< 5 , 5-6, 6-8, 8-10, 10-20, 20-30, 30-40, 40-45, and 45-55), quadratic trends in average cohort education, quadratic trends in county-level demographics, and industry-year fixed effects. The table reports the nine-year sum of event-time coefficients. The regressions are weighted by the 1990 cohort size. Standard errors are calculated using the delta method and are clustered by commuting zone. *** **, * denotes statistical significance at the 1%, 5%, and 10% levels, respectively.

Source: LEHD.

Table V: effects across the preregulatory earnings distribution.

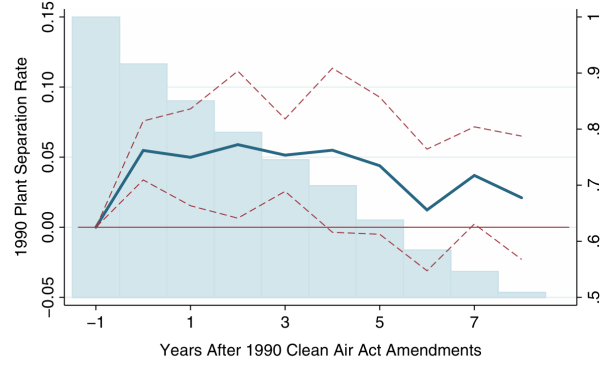


FIGURE V

Figure V: job transition rates from the newly regulated sector.

6.6 Figure VI: where do workers go?

Read:

- Figure VI decomposes separation rates by whether workers move across industries and/or counties.
- The clearest increase is movement to a different industry within the same county.
- Cross-county movement is less central than cross-industry movement.
- This implies that adjustment often happens within the local labor market rather than through migration.

Economic message: Sectoral reallocation is mainly occupational/industry reallocation inside local labor markets, not mass geographic migration.

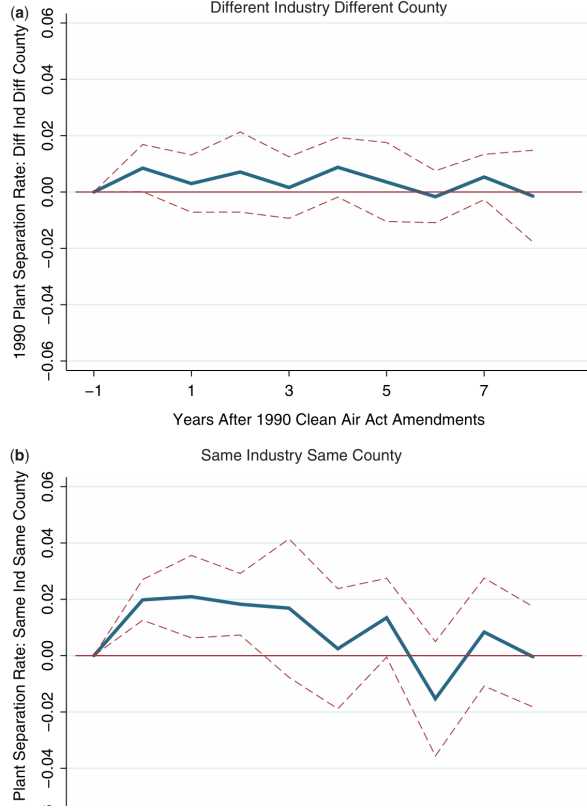


Figure VI, part 1: transition destinations A and B.

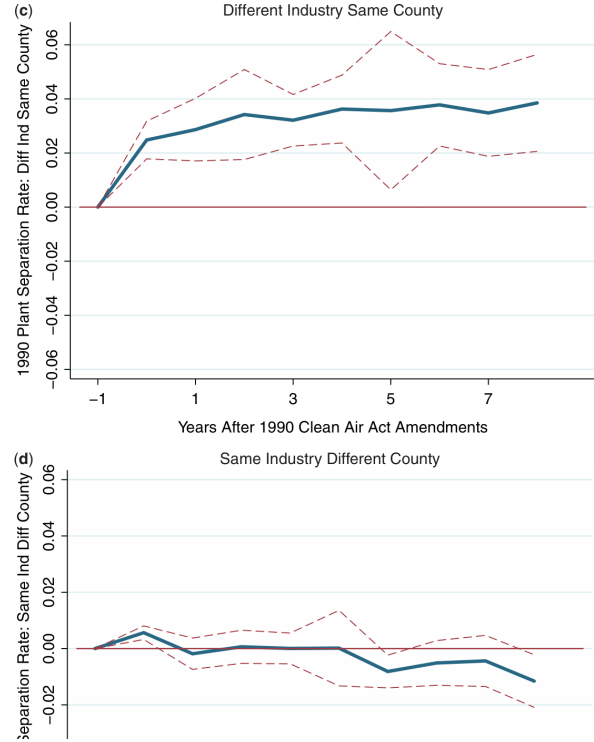


Figure VI, part 2: transition destinations C and D.

7 Short summary

- The paper estimates the worker-level transitional costs of the 1990 Clean Air Act Amendments.
- The empirical design uses county, pollutant, plant, and time variation in regulation.
- The main data advantage is linked worker-firm LEHD data, which follows workers after they leave regulated plants.
- Newly regulated sectors lose employment after the policy shock.
- Workers initially employed in treated sectors lose about 20% of a preregulatory year's earnings in present discounted value over nine years.
- The aggregate forgone wage bill is about \$5.4 billion in 1990 dollars.
- Stayers experience little loss; separators experience large losses.
- Different-industry transitions are especially costly, consistent with industry-specific human capital.
- Losses are larger for older workers, women, higher-earnings workers, workers in larger plants, and workers in high-unemployment counties.
- The worker costs are meaningful but small relative to estimated health benefits of the 1990 CAAA.

8 Conclusion

8.1 Core conclusion

Walker shows that environmental regulation can generate significant transitional costs for workers in newly regulated sectors. The average affected worker suffers a persistent but partially recovering earnings loss. These costs are economically important for the workers who bear them.

8.2 Economic intuition

Regulation raises production costs for polluting plants in nonattainment counties. Employment shifts away from those plants. Workers who remain largely keep their wages, but displaced workers must search for new jobs, often in different industries. Because skills, wage premia, and job matches are partly sector-specific, the move is costly.

8.3 Incremental contribution revisited

The paper changes the unit of analysis from the sector to the worker. This makes the true incidence of reallocation visible: the same sectoral employment change can have small costs if workers move smoothly, or large costs if workers experience nonemployment and lower future wages.

8.4 Implications

Policy evaluation should distinguish three objects:

- the aggregate efficiency and health benefits of environmental policy;
- the compliance and employment costs borne by firms and sectors;
- the distributional and transitional costs borne by workers who must move.

The paper's estimates support the view that the 1990 CAAA had large net benefits but also created concentrated losses for a specific group of workers.

8.5 Limitations

The analysis is partial equilibrium and focused on directly exposed workers. It does not measure all possible labor-market effects, such as workers who would have entered the regulated sector but did not, or workers in sectors that gained from pollution-control demand. The LEHD sample is also based on states with data before 1991, so national extrapolation requires assumptions.

8.6 Takeaway

The enduring takeaway is that sectoral reallocation is not frictionless. Even when a policy has large social benefits, workers displaced from affected sectors can face substantial short- and medium-run earnings losses. Good policy analysis should measure those costs directly and consider whether adjustment assistance or insurance is warranted.